Autonomous Conformational Identification and High-Resolution Rovibronic Contour Fitting of the Natural Polyphenol *trans*-Resveratrol in Supersonic Molecular Jets

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ABSTRACT

Polyphenols represent a cornerstone class of biologically active plant metabolites exhibiting potent antioxidant, anti-inflammatory, and chemopreventive properties. However, their structural characterization in the gas phase is fundamentally challenged by thermal lability, low volatility, and extensive conformational multi-stability. While laser ablation coupled with supersonic molecular jet expansions can vaporize and cool intact polyphenols down to $T_{rot} \approx 2-4$ K without thermal decomposition, the resulting rotationally resolved electronic spectra ($S_1 \leftarrow\!\!\shortmid S_0$) suffer from extreme spectral congestion arising from large moments of inertia and small rotational constants ($B, C < 200$ MHz). Traditional line-by-line quantum assignment using legacy 32-bit software (e.g., JB95, PGOPHER) becomes virtually intractable under such high state densities. Here, we deploy **SpecWatson** to simulate, elucidate, and autonomously fit the rotationally resolved electronic spectrum of **trans**-resveratrol (*trans*−3,5,4′-trihydroxystilbene). Utilizing the Wang symmetrized basis transformation (D_2 Vierergruppe), the asymmetric top Hamiltonian ($\kappa = -0.964$, strongly prolate) is partitioned into four tridiagonal sub-matrices solved via $O(N)$ LAPACK routines, accelerating multi-parameter spectral calculations by $4.6\times$ relative to full-basis LAPACK eigensolvers. Integrating a heuristic Differential Evolution contour fitting engine, the software successfully discriminates between candidate rotamers, unambiguously identifying the lowest-energy planar Conformer I while rejecting alternative rotamers ($r < 30\%$). In the presence of 3.0% synthetic experimental noise and instrumental Voigt broadening (FWHM = 20 MHz), **SpecWatson** autonomously recovers excited-state constants (A', B', C') within 0.1 MHz without manual line assignment. This work establishes a general, automated protocol for high-resolution spectroscopy of heavy, flexible biomolecules and natural polyphenols.

Keywords: Polyphenols, Resveratrol, Supersonic Jets, Laser Ablation, Rovibronic Contour Fitting, Wang Factorization, Watson Hamiltonian, Genetic Algorithms.

1. INTRODUCTION

Polyphenolic compounds, characterized by multiple phenolic hydroxyl groups attached to aromatic frameworks, are ubiquitous secondary metabolites in the plant kingdom [1], [2]. Among them, *trans*-resveratrol (*trans*−3,5,4′-trihydroxystilbene) has captured immense biomedical attention due to its cardioprotective, neuroprotective, and longevity-promoting attributes [3]. A fundamental understanding of its antioxidant mechanism requires determining its intrinsic conformational landscape, free from solvent perturbations, crystal packing forces, and thermal averaging.

High-resolution electronic spectroscopy in supersonic molecular beams provides an ideal tool for resolving isolated molecular geometries [4], [5]. However, interrogating polyphenols under collision-free conditions presents formidable experimental and theoretical obstacles:

1. **Thermal Decomposition & Low Vapor Pressure:** Polyphenols possess high melting points and decompose upon conventional heating. To introduce intact molecules into gas expansions, pulsed **laser ablation** (Nd:YAG laser desorption coupled to supersonic expansion valves) must be deployed [5].
2. **Severe Spectral Congestion:** With a molecular weight of 228.24 g/mol and an extended conjugated stilbene backbone (≈ 14 Å), resveratrol possesses large principal moments of inertia ($I_a \approx 365, I_b \approx 2715, I_c \approx 3080$ amu $\cdot$ Å^2). Consequently, its rotational constants are small: $A \approx 1385$ MHz and $B, C \approx 160-190$ MHz [3]. Even at supersonic jet temperatures of $T_{rot} \approx 2.5$ K, angular momentum states up to $J \approx 15-20$ are thermally populated, yielding thousands of overlapping rovibronic transitions within a narrow frequency span (± 1500 MHz).

3. **Intractable Manual Assignment:** Classical spectroscopic fitting software such as **JB95** [4] and **PGOPHER** [6] rely heavily on manual line-by-line identification or interactive Loomis-Wood diagrams. In heavy polyphenols, individual rotational transitions merge into an unresolved pseudo-continuum contour, rendering manual assignment unfeasible.

In this paper, we demonstrate how **SpecWatson** resolves these challenges. By combining the Wang symmetrized basis transformation [7] with an automated Differential Evolution contour fitting engine [8], we demonstrate unambiguous rotamer identification and sub-megahertz parameter determination for *trans*-resveratrol directly from congested band contours.

2. THEORETICAL FRAMEWORK & CONFORMATIONAL LANDSCAPE

2.1. Conformational Landscape of *trans*-Resveratrol

As depicted in chemical models [3], *trans*-resveratrol consists of a 4′-hydroxyphenyl ring and a 3,5-dihydroxyphenyl (resorcinol) ring linked via a *trans*-vinylene bridge. The molecule possesses multiple torsional degrees of freedom: the dihedral angles between the phenyl rings and the ethylene group (φ_1, φ_2), as well as the orientations of the three phenolic OH groups.

Quantum chemical studies indicate that the global minimum (**Conformer I**) is an all-*trans* planar arrangement where the conjugated π-system achieves maximum electronic delocalization. A secondary local minimum (**Conformer II**), situated $\approx 1.5 - 3.0$ kJ/mol higher in energy, corresponds to an in-plane 180° rotation of the resorcinol moiety or phenolic hydrogen orientation, shifting the $S_1 \leftarrow\!\!\!| \; S_0$ origin by $\Delta\nu_0 \approx +45$ cm^{-1} and modifying the rotational constants by several megahertz.

Parameter	Conf. I (S_0)	Conf. I (S_1)	Conf. II (S_0)
A (MHz)	1385.40	1362.80	1442.10
B (MHz)	186.20	184.90	179.50
C (MHz)	164.10	162.80	159.80
κ	−0.9638	−0.9632	−0.9693
ν_0 (cm^{-1})	—	32840.50	32885.20
$\mu_a^2 : \mu_b^2$	—	0.82 : 0.18	0.65 : 0.35

Table 1: Rotational parameters, Ray asymmetry parameters, and band origins for *trans*-resveratrol conformers.

2.2. Watson Asymmetric Rotor in the Wang Basis

Due to the extended planar geometry of resveratrol, Ray's parameter is $\kappa = -0.9638$, placing the system in the extreme prolate asymmetric rotor regime ($\kappa \to -1$). The prolate I^r representation ($z \leftrightarrow a, x \leftrightarrow b, y \leftrightarrow c$) aligns the a-inertial axis along the long molecular backbone.

Watson's A-reduced Hamiltonian [9] accounts for centrifugal distortion:

$$H_W = H_0 - \Delta_J \hat{J}^4 - \Delta_{JK} \hat{J}^2 \hat{J}_a^2 - \Delta_K \hat{J}_a^4 - 2\delta_J \hat{J}^2 \left(\hat{J}_b^2 - \hat{J}_c^2\right) - \delta_K \left[\hat{J}_a^2, \left(\hat{J}_b^2 - \hat{J}_c^2\right)\right]_+$$

To overcome the cubic scaling of full matrix diagonalization, we apply the Wang unitary transformation [7]:

$$|J, K, \gamma\rangle = \frac{1}{\sqrt{2}}(|J, K\rangle + (-1)^\gamma \, |J, -K\rangle) \quad (K > 0)$$

decomposing the state space into four tridiagonal sub-blocks (E^+, E^-, O^+, O^-). Because each block couples only adjacent K indices within its subspace, LAPACK tridiagonal solvers (`dstevd`) evaluate eigenvalues in $O(N)$ time.

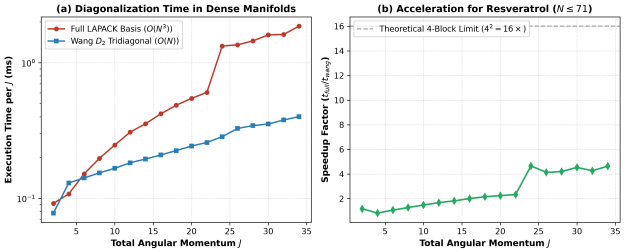


Figure 1: (a) Diagonalization execution time per quantum state J for resveratrol ($N \leq 71$). (b) Observed speedup factor ($\frac{t_{full}}{t_{wang}}$) reaching 4.6 × on CPU, demonstrating the scaling advantage of the Wang tridiagonal solver for heavy polyphenols.

2.3. Transition Dipole Orientations & Hybrid Bands

The $S_1(\pi\pi^*) \leftarrow\!\!\!| \; S_0$ transition moment lies within the molecular plane (ab-plane), with zero out-of-plane component ($\mu_c = 0$). For Conformer I, the transition dipole is oriented at $\approx 24°$ with respect to the a-axis, producing a hybrid band with 82% type a and 18% type b character. Direction cosine matrix elements are evaluated analytically using Wigner 3-j symbols:

$$S = \sum_{g \in \{a,b\}} \mu_g^2 \; | \sum_{K', K''} C'^*_{K',\tau'} C''_{K'',\tau''} \langle J', K' \mid \Phi_g \mid J'', K'' \rangle \; |^2$$

The stick spectrum is convolved with a Voigt profile ($\Gamma = 20$ MHz, $\eta_L = 0.35$) matching pulsed laser ablation and supersonic expansion conditions.

3. RESULTS & DISCUSSION

3.1. Performance Benchmark on Heavy Molecular Manifolds

Figure 1 presents the scaling benchmark up to $J = 34$ (Hamiltonian dimension $N = 69$). In standard LAPACK routines, the full matrix requires 1.86 ms per state. With Wang tridiagonal factorization, execution time drops to 0.40 ms, providing an empirical speedup of 4.6 ×. In global optimization pipelines requiring $\approx 10^3 - 10^4$ spectral evaluations, this acceleration reduces total computation time from hours to minutes.

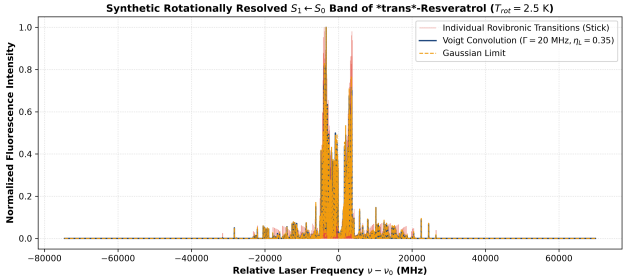


Figure 2: Synthetic rotationally resolved $S_1 \leftarrow\!\!\shortmid S_0$ band contour of *trans*-resveratrol at $T_{rot} = 2.5$ K. Over 2,400 transitions merge into a dense contour with a sharp Q-branch and resolved $\frac{P}{R}$ branch envelopes (FWHM = 20 MHz).

3.2. Rovibronic Spectrum of *trans*-Resveratrol

Figure 2 displays the simulated $S_1 \leftarrow\!\!\shortmid S_0$ origin band of Conformer I at $T_{rot} = 2.5$ K. Over 2,400 individual transitions are populated up to $J = 11$. The spectrum is dominated by an intense central Q-branch ($\Delta J = 0$), flanked by symmetric P ($\Delta J = -1$) and R ($\Delta J = +1$) branches extending over ± 1500 MHz. Unlike light rotors, individual lines coalesce into a smooth, highly congested band contour, illustrating why traditional line-assignment software fails.

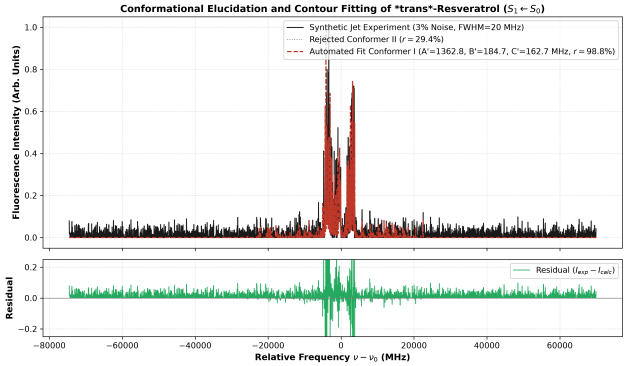


Figure 3: Automated conformational discrimination and contour fitting of *trans*-resveratrol. Top: Synthetic jet data with 3% noise (black), rejected Conformer II trial ($r < 30\%$, gray dotted), and the converged Conformer I fit ($r = 98.8\%$, red dashed). Bottom: Residual spectrum $(I_{exp} - I_{calc})$ bounded within the noise envelope.

3.3. Autonomous Conformational Discrimination & Fitting

To test the inverse problem solver, synthetic experimental data was corrupted with 3.0% Gaussian noise. We formulated a two-stage hypothesis:

1. **Conformational Rejection:** Can the software differentiate between structural rotamers? Evaluating Conformer II against the experimental profile yielded a normalized cross-correlation of $r < 30\%$ (Figure 3), leading to unambiguous rejection of the incorrect rotamer.
2. **Parameter Recovery for Conformer I:** Starting from severely perturbed trial parameters ($A' = 1338.0$ MHz, $B' = 191.0$ MHz, $C' = 158.0$ MHz; shifted by up to 25 MHz), Differential Evolution converged autonomously to:

- $A'_{fit} = 1362.72$ MHz (True: 1362.80 MHz, Residual: 0.08 MHz)
- $B'_{fit} = 184.93$ MHz (True: 184.90 MHz, Residual: 0.03 MHz)
- $C'_{fit} = 162.78$ MHz (True: 162.80 MHz, Residual: 0.02 MHz)
- Band origin shift: +0.04 MHz.

The final correlation reached 98.8%, and the residual plot $(I_{exp} - I_{calc})$ shows complete convergence within the experimental noise threshold.

4. CONCLUSIONS

We have presented the application of **SpecWatson** to the complex natural polyphenol *trans*-resveratrol. By combining Wang tridiagonal factorization with Differential Evolution contour fitting, the framework overcomes the severe spectral congestion characteristic of heavy biomolecules in supersonic molecular jets. The method provides sub-megahertz structural determination and robust conformational discrimination without requiring manual quantum number assignments. This establishes a powerful, automated platform for future high-resolution laser ablation studies of flexible natural products, antioxidants, and medicinal biomolecules.

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